

Project Design Phase

Solution Architecture

Date	13-07-2026
Team ID	SWTID-2026-7371
Project Name	House Rent
Maximum Marks	4 Marks

Solution Architecture:

The **House Rent Management System** follows the **MERN Stack Architecture**, which integrates **React.js**, **Node.js**, **Express.js**, and **MongoDB Atlas** to provide a secure and efficient rental property management platform.

The solution architecture bridges the gap between user requirements and technical implementation by providing a structured flow of data between the frontend, backend, and database.

The main goals of the solution architecture are:

- Provide a secure platform for user registration and login using **JWT Authentication**.
- Allow users to browse, search, view, and book rental properties through a responsive React.js interface.
- Process user requests using RESTful APIs developed with Express.js.
- Store and manage user, property, and booking information securely in MongoDB Atlas.
- Support scalable and modular development using the MERN Stack architecture for future enhancements.

Example - Solution Architecture Diagram:

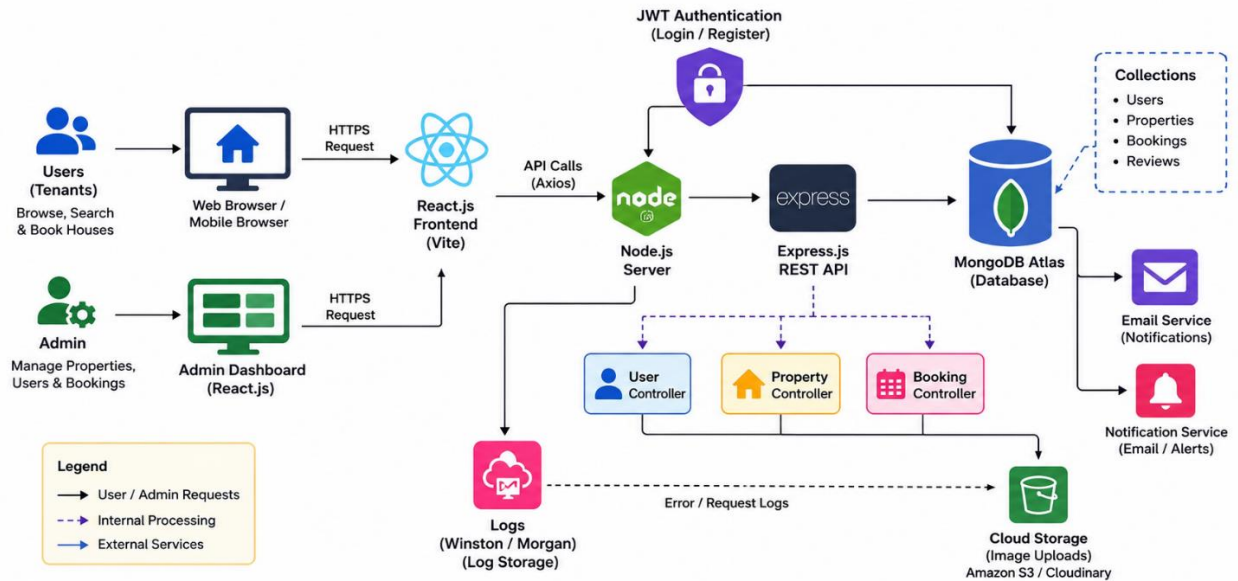


Figure 1: Architecture and data flow of the House Rent Management System

Architecture Workflow:

1. The user accesses the application through the React.js frontend.
2. The frontend sends API requests to the Express.js backend using Axios.
3. The backend validates requests and authenticates users using JWT.
4. Controllers process the business logic for users, properties, and bookings.
5. Mongoose communicates with MongoDB Atlas to perform CRUD operations.
6. The backend returns the requested data to the frontend.
7. The frontend displays property details, booking information, and user data to the user.

Summary:

The House Rent Management System follows a modular MERN Stack architecture that ensures secure authentication, efficient data management, and seamless communication between the frontend, backend, and MongoDB Atlas database. This architecture improves scalability, maintainability, and overall application performance.